25901231 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEECT202 - DATA SCIENCE FOR AI

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Identify the main benefit of moving tensors to GPU. [K] / [CO1]

   a). Reduced storage b). Smaller tensor size

   c). Improved naming d). Faster parallel computation

2. Distinguish between 1D convolution and 2D convolution based on application. [U] / [CO2]

   a). 1D is used for images, 2D for audio b). 2D convolution works only on text

   c). 1D is used for sequential data, 2D is used for images d). Both are used only in CNN outputs

3. Identify the problem that occurs when gradients shrink during Backpropagation Through Time over long sequences. [U] / [CO3]

   a). Overfitting b). Vanishing gradient problem

c). Exploding gradient problem d). Underfitting

4. Identify effect of no causal mask? [U] / [CO4]

a). Leaks future info b). Faster training

c). Better parallelism d). Smaller model

5. Solve for the output size when a 32×32 image, filter size 5×5, stride = 1, and zero padding = 2 are applied. [A] / [CO2]

a). 28 × 28 b). 30 × 30

c). 32 × 32 d). 34 × 34

6. Define floating point numbers in tensor computation. [K] / [CO1]

a). Integers only b). Numbers with fractional parts

c). Boolean values d). Character data

7. Demonstrate understanding of max pooling by selecting the correct output.For a window 2,5,1,3 max pooling gives: [U] / [CO2]

a). 2 b). 3

c). 5 d). 1

8. Select the reason why Recurrent Neural Networks are suitable for language modeling tasks. [U] / [CO3]

a). They process input in parallel b). They retain contextual information from earlier words

c). They ignore previous inputs d). They require less training data

9. Translate the need for attention in NLP tasks? [U] / [CO4]

a). To forget context b). To shorten inputs

c). To align source and target parts in translation d). To add noise

10. List primary use case for cross-attention? [K] / [CO4]

a). Intra-sequence relations b). Image classification

c). Encoder-decoder sequence tasks

d). Single token processing

11. Explain how named tensors reduce programming errors. [U] / [CO1]

a). By increasing speed

b). By improving readability

c). By reducing memory

d). By compressing data

12. Classify padding types used in sizing patterns. [A] / [CO2]

a). Zero and valid

b). Same and valid

c). Max and average

d). Sparse and dense

13. Identify the primary goal of generative machine learning models. [U] / [CO3]

a). Learn decision boundaries only

b). Minimize classification error

c). Generate new data similar to training data

d). Perform feature selection

14. Classify self-attention as intra-sequence mechanism? [A] / [CO4]

a). Decoder-exclusive

b). Yes, within one sequence

c). No QKV used

d). Cross-sequence only

15. Identify which component in an LSTM stores information for long periods? [U] / [CO3]

a). Memory cell

b). Hidden state

c). Input gate

d). Output gate

16. Define tensor storage. [K] / [CO1]

a). Tensor shape

b). Tensor name

c). Tensor metadata

d). Physical memory where tensor data is stored

[1x16 = 16]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

17. Explain the role of stride in memory-efficient tensor operations. [U] / [CO1]

18. Comprehend the concept of local connectivity in convolutional neural networks. [U] / [CO2]

19. Explain the concept of Recurrent Neural Networks (RNN) and distinguish them from Feed Forward Neural Networks. [U] / [CO3]

20. Describe why scaling by sqrt(d_k) is used in attention overview. [U] / [CO4]

21. Explain the role of channels in image tensors. [U] / [CO1]

22. Explain how convolutional layers extract features from images. [U] / [CO2]

23. Explain many-to-one and many-to-many types of RNN [U] / [CO3]

24. Explain step-by-step: from QKV projection to weighted sum in attention. [U] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **12** marks

25. Describe the role of volumetric data in modern machine learning applications. [U] / [CO1]

26. Explain in detail the common layer patterns used in CNN architectures with suitable diagrams. [U] / [CO2]

27. Describe the structure of an autoencoder and summarize its role in feature learning. [U] / [CO3]

28. Explain how global and local attention are combined in long text processing. [U] / [CO4]

[12x2 = 24]